

PYTHON FOR BEGINNERS AND PROFESSIONALS

SESSION 02

Escape Sequences and Operators

Escape Sequences

An escape sequence is a special character that starts with a backslash `\`. It lets you put things inside a string that are otherwise hard to type directly.

Escape Sequence	What it does	Example
<code>\n</code>	New line	"Hello\nWorld"
<code>\t</code>	Tab space	"Name:\tRahul"
<code>\\</code>	Prints a backslash	"C:\\Users\\Rahul"
<code>\"</code>	Prints a double quote	"He said \"Hello\""
<code>\'</code>	Prints a single quote	'It\'s Python'

Arithmetic Operators

Arithmetic operators are used to perform basic mathematical operations.

Operator	Name	Example	Result
+	Addition	10 + 3	13
-	Subtraction	10 - 3	7
*	Multiplication	10 * 3	30
/	Division	10 / 3	3.333...
//	Floor Division	10 // 3	3
%	Modulus	10 % 3	1
**	Exponentiation	2 ** 4	16

Assignment Operators

Assignment operators are shortcuts for updating a variable's value. Instead of writing $x = x + 5$ you can write $x += 5$.

Operator	Same as	Example
+=	$x = x + n$	$x += 5$
-=	$x = x - n$	$x -= 3$
*=	$x = x * n$	$x *= 2$
/=	$x = x / n$	$x /= 4$
//=	$x = x // n$	$x //= 3$
%=	$x = x \% n$	$x \% = 3$
**=	$x = x ** n$	$x ** = 2$

Operator Precedence

When multiple operators are in one expression, Python follows a specific order – just like BODMAS in maths.

Order (highest to lowest):

****** → Exponentiation

***, /, //, %** → Multiplication & Division

+, - → Addition & Subtraction

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# Without knowing precedence, this looks confusing
print(2 + 3 * 4)          # 14, NOT 20 (multiplication first)
print(10 - 2 ** 3)        # 2,  NOT 512 (exponent first)
print(10 // 2 + 3)        # 8,  NOT 1

# Use parentheses to force the order you want
print((2 + 3) * 4)         # 20
print((10 - 2) ** 3)      # 512
```

Comparison Operators

Comparison operators compare two values and always return a boolean – either **True** or **False**.

Operator	Meaning	Example	Result
=	Equal to	5 == 5	TRUE
!=	Not equal to	5 != 3	TRUE
>	Greater than	7 > 3	TRUE
<	Less than	2 < 1	FALSE
>=	Greater than or equal	5 >= 5	TRUE
<=	Less than or equal	4 <= 3	FALSE

Logical Operators

Logical operators are used to combine multiple conditions together.

Operator	Meaning	Returns True when
and	Both conditions	Both are True
or	Either condition	At least one is True
not	Opposite	Original is False

Some more Operators

These operators requires more python knowledge so we will be covering them in future sessions.

- Identity Operators
- Membership Operators
- Bitwise Operators

Practice Questions

Q1: Take two numbers as input from the user. Print their sum, difference, product, and remainder.

Q2: Take a number as input. Print whether it is even or odd using the % operator and a comparison operator.

Q3: Take the user's age as input. Check and print whether they are eligible to vote (age ≥ 18) and whether they are a senior citizen (age ≥ 60). Print both results.

Q4: A student scored marks in 3 subjects. Take all three as input, calculate the total and average, and print both using an f-string.

Practice Questions (Homework)

Q5: Take a number as input. Print the result of that number raised to the power of 3 using `**`. Also print what `// 7` and `% 7` give for the same number.

Q6: Take two numbers as input. Without using `*`, calculate and print their product using `+=` in a way that adds the first number to itself the second number of times. (Think carefully.)

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